



MATLAB for Research

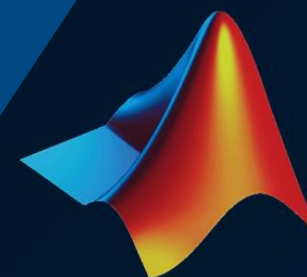
Computing, data analysis, visualization, & mapping



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MATLAB with

Overarching Agenda

10:00 AM – 10:45 AM (MDT)

- **MATLAB for Research:** low-code data analysis, latest updates, interoperability

11:00 AM – 12:00 PM (MDT)

- **Agentic AI Workflows:** available tools, leveraging MATLAB, and accelerating research workflows

Slides & File access: <https://github.com/drLKeen>

Your MathWorks Team



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Seminar Feedback

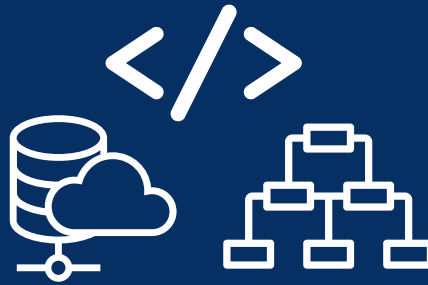
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Experience – Fill out form



Ease of Use



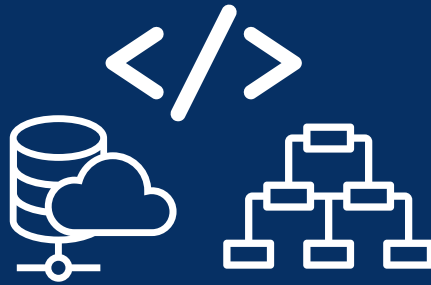
Software Workflows



Integrations



Ease of Use



Software Workflows



Integrations

See all the details in the Release Notes

Filter results by:

- Release range
- Category
- Text filter
- Incompatibilities
- Highlights

The screenshot shows the MATLAB Release Notes page for R2026a. The page title is "MATLAB Release Notes" with a sub-header "R2026a". There are links for "Bug Reports" and "Bug Fixes". A "YOUR SELECTIONS" bar shows "Performance" selected. A yellow box highlights "Found 81 notes". The "Release Range" is set to "R2023a to R2026a". A dropdown menu for "Performance" is open, showing a list of categories: Environment, Language and Programming, Data Analysis, Data Import and Export, Mathematics, Graphics, App Building, Performance (checked), Software Development Tools, External Language Interfaces, and Hardware Support. The "Text Filter" is set to "MATLAB Release Notes". The "Sort Release" is set to "Latest to Earliest". The "Performance" section is expanded, showing "duration Data Type: Improved performance". A list of improvements is shown: Arithmetic operations, Array creation, Array indexing, Concatenation, Reshaping, and Sorting. A yellow box highlights the text "about 108x faster than in the previous release." Below this, a code block shows a function for timing tests. A black box with a white URL is overlaid on the bottom right of the code block. The URL is <https://www.mathworks.com/help/matlab/release-notes.html>. The page also shows execution times for R2025b (12.93 s) and R2026a (0.12 s).

MATLAB Release Notes R2026a expand all in page

Bug Reports | Bug Fixes

YOUR SELECTIONS Performance x

Found 81 notes Release Range: R2023a to R2026a Share

R2023a to R2026a Performance Incompatibilities Highlights Sort Release: Latest to Earliest

Text Filter: MATLAB Release Notes

▼ R2026a New Features, Bug Fixes, Compatibility Updates

Performance

▼ duration Data Type: Improved performance

Operations on duration arrays supported

- Arithmetic operations
- Array creation
- Array indexing
- Concatenation
- Reshaping
- Sorting

For example, this code creates a duration scalar. The code is about 108x faster than in the previous release.

```
function timingTest
for i = 1:1e6
    d = duration(0,0,30);
end
end
```

<https://www.mathworks.com/help/matlab/release-notes.html>

The approximate execution times are:

R2025b: 12.93 s

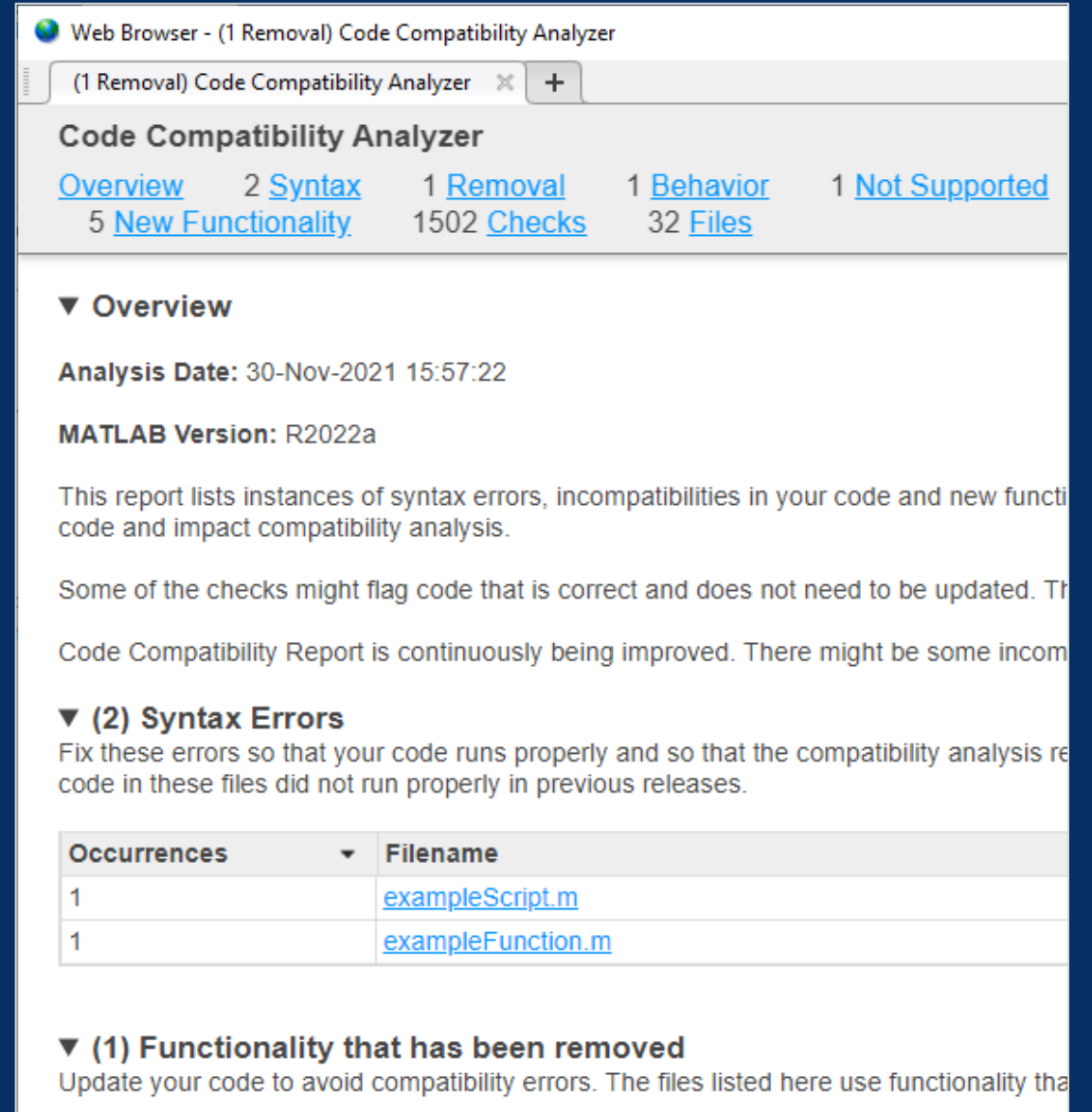
R2026a: 0.12 s

Make upgrades easier

Code Compatibility Analyzer App

Additional resources:

- [Upgrade strategies white paper](#)
- [Software Upgrade Consulting Service](#)



The screenshot shows a web browser window titled "Web Browser - (1 Removal) Code Compatibility Analyzer". The address bar shows the URL "http://localhost:10000/(1 Removal) Code Compatibility Analyzer". The page title is "Code Compatibility Analyzer". The navigation bar includes links: [Overview](#), [2 Syntax](#), [1 Removal](#), [1 Behavior](#), and [1 Not Supported](#). Below the navigation bar, there are counts: [5 New Functionality](#), [1502 Checks](#), and [32 Files](#).

▼ Overview

Analysis Date: 30-Nov-2021 15:57:22

MATLAB Version: R2022a

This report lists instances of syntax errors, incompatibilities in your code and new functionality and impact compatibility analysis.

Some of the checks might flag code that is correct and does not need to be updated. The Code Compatibility Report is continuously being improved. There might be some incompatibilities.

▼ (2) Syntax Errors

Fix these errors so that your code runs properly and so that the compatibility analysis report code in these files did not run properly in previous releases.

Occurrences	Filename
1	exampleScript.m
1	exampleFunction.m

▼ (1) Functionality that has been removed

Update your code to avoid compatibility errors. The files listed here use functionality that has been removed.

Code Compatibility Analyzer

Live Editor

Make scripts readable

- Richly formatted text
- Interleave code & output

Add interactivity

- Hyperlinks and table of contents
- Live Controls to adjust parameters

Share with non-MATLAB users

- Export as report (pdf, docx, etc.)
- Convert to Jupyter notebook
- Share through GitHub and MATLAB Online

SunriseSunset.mlx x +

Estimating Sunrise and Sunset

We can estimate sunrise and sunset times if we know the latitude, longitude, and UTC offset. The [solar declination](#) (δ) is the angle of the sun relative to the earth's equatorial plane. On any given day of the year (d), using the latitude (ϕ), the sun's declination (δ) and the solar time correction (SC) we can calculate sunrise and sunset times.

Solar Declination	Sunrise	Sunset
$\delta = \sin^{-1} \left[\sin(23.45) \sin \left(\frac{360}{365} (d - 81) \right) \right]$	$12 - \frac{\cos^{-1}(-\tan \phi \tan \delta)}{15^\circ} - \frac{SC}{60}$	$12 + \frac{\cos^{-1}(-\tan \phi \tan \delta)}{15^\circ} - \frac{SC}{60}$

Table of Contents

- [Estimating the Sunrise and Sunset Times](#)
- [Plot Yearly Results](#)

Estimating the Sunrise and Sunset Times

Set the latitude, longitude, and UT offset.

1

lat = 42

2

lon = -71

3

UTCoff = -5

Estimate the sunrise and sunset times. We use the custom equationOfTime function to calculate the solar time [correction](#) (SC).

4

5

6

```
day = 1:365;
timeCorr = equationOfTime(day)

timeCorr = 1x365
-3.7052 -4.1497 -4.5894 -5.0239 -5.4528 -5.8756 -6.2921 -6.7018 ...

plotSunriseSunset(lon, UTCoff, timeCorr, day, lat)
```

Sunrise and Sunset

 MathWorks®

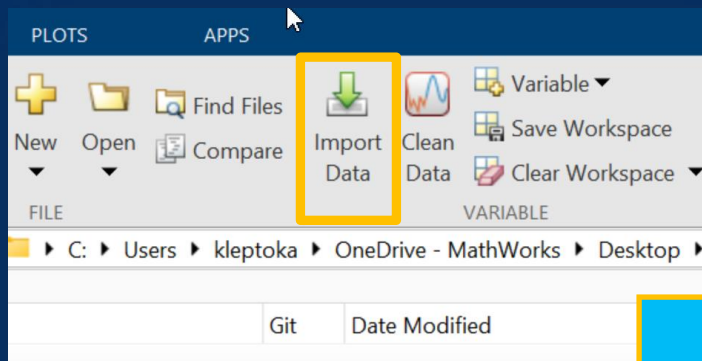
9

Data Handling in MATLAB:

Import Data Live Task

supports:

- *NetCDF (R2023b)*
- *HDF5 (R2024a)*
- *GRIB (R2024b)*
- *And more*



Code
Generated

A screenshot of the MATLAB 'Import Data' task window. The window shows the selection of an HDF5 file from a local path. The 'Select data to import' section shows a tree view of the file's contents, with 'DS_UTCTime_1' selected. The 'Specify subsetting options for datasets' section is expanded, showing options for code generation. The 'Display results' section shows the generated MATLAB code.

Import Data ▶ ✓ Autorun ? ⋮

GLAH06_634_2103_002_0196_2_01_0001 = Struct of HDF5 data imported from GLAH06_634_2103_002_0196_2_01_0001.H5

▼ **Select source**

File: C:\Users\oghosh\Downloads\GLAH06_634_2103_002_0196_2_01_0001. Browse...

Type: HDF5, Size: 4.45 MB

▼ **Select data to import**

Search

- Attributes (83)
- Groups (5)
 - ANCILLARY_DATA
 - BROWSE
 - Data_1HZ
 - Attributes (1)
 - Datasets (1)
 - DS_UTCTime_1
 - ☒ DS_UTCTime_1
 - Attributes (9)
 - Groups (8)
 - Data_40HZ
 - METADATA
 - Attributes (3)
 - description
 - HDFVersion

/Data_1HZ/DS_UTCTime_1

Size: 1332
MaxSize: Inf
Datatype: H5T_IEEE_F64LE (double)
ChunkSize: 38400
Filters: deflate(6)
FillValue: 0.000000

▼ **Specify subsetting options for datasets**

▼ **Specify code generation type**

▼ **Display results**

▼ Hide code

```
% Create a structure to store imported HDF5 data
GLAH06_634_2103_002_0196_2_01_0001 = struct();

filename = "C:\Users\oghosh\Downloads\GLAH06_634_2103_002_0196_2_01_0001.H5";

GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Name = "Data_1HZ";

GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Datasets(1).Name = "DS_UTCTime_1";
GLAH06_634_2103_002_0196_2_01_0001.Groups(1).Datasets(1).Value = h5read(filename, "/Data_1HZ/DS_UTCTime_1");

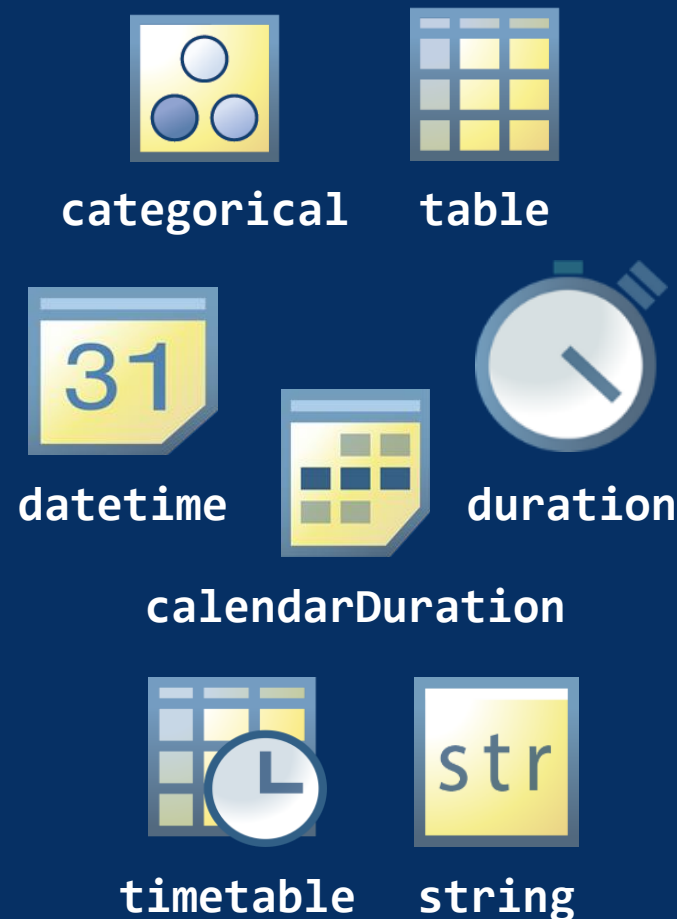
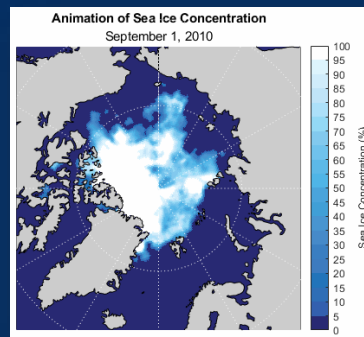
clear filename

% Display results
GLAH06_634_2103_002_0196_2_01_0001
```

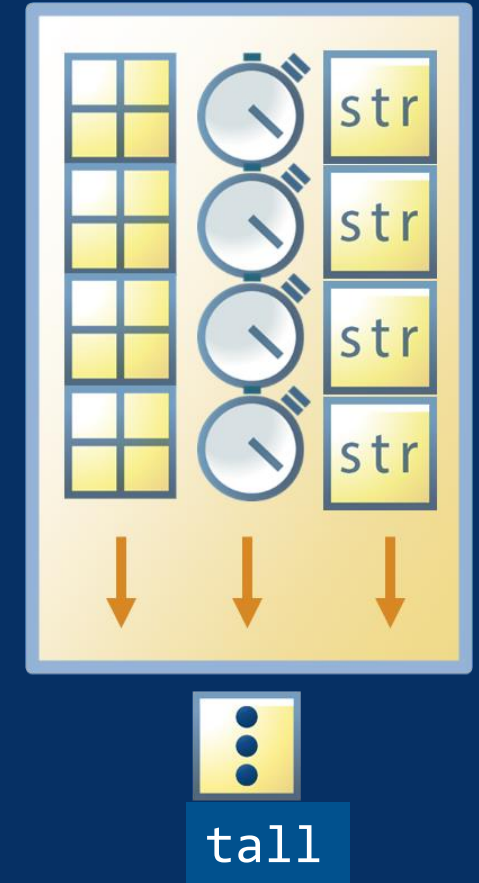
Supported data formats and types in MATLAB

- **Raster file formats:** GeoTIFF, USGS DEM, DEM, DTED, Arc ASCII Grid, GTOPO30, ETOPO, and worldfile
- **Vector file formats:** ESRI shapefiles, KML, GPX, VMAP0, GSHHS, and .gpkg
- **Image file formats:** TIFF, JPEG, PNG, JPEG2000, LandSat, NITF and HDR
- **Scientific data:** NetCDF, HDF5, GRIB, CDF, FITS, and multiband files (BIP, BIL, BSQ)
- Community Tools to work SAC and SEG-Y

[Data Formats >>](#)



[Data Format Functions >>](#)



[Tall Arrays >>](#)

Where are you from?



Answer: Where you're from depends on the context

Country

State

Major City

Suburb

I'm from the U.S.



I'm from Massachusetts



I'm from **Boston**

42.361145° N, -71.057083° W

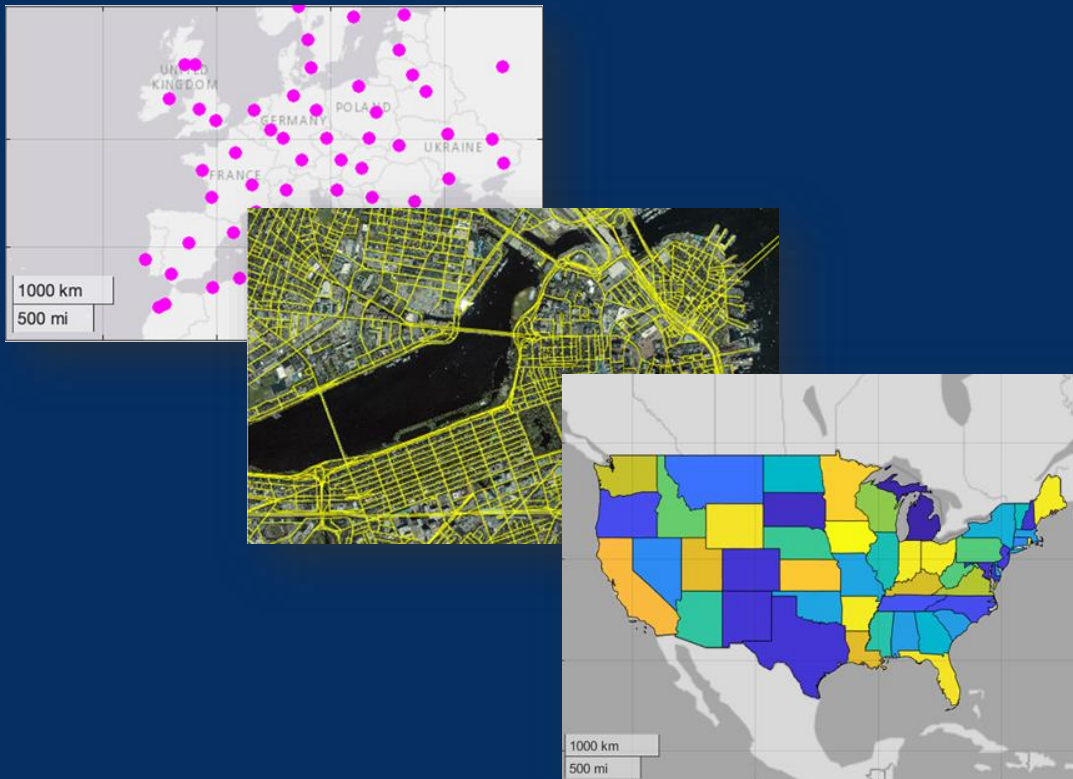


I'm from "**Boston**"
(suburbs)

Unified file readers

Vector data

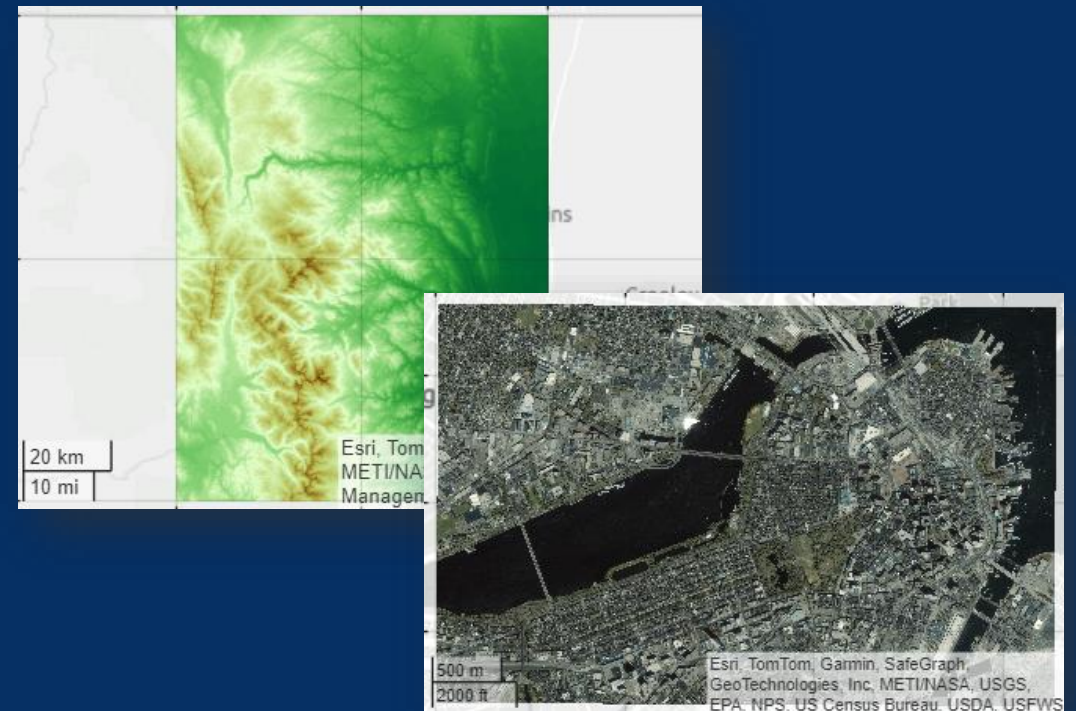
(Geometric features - points, lines, polygons)



```
T = readgeotable(filename);
```

Raster data

(Grid-based data)

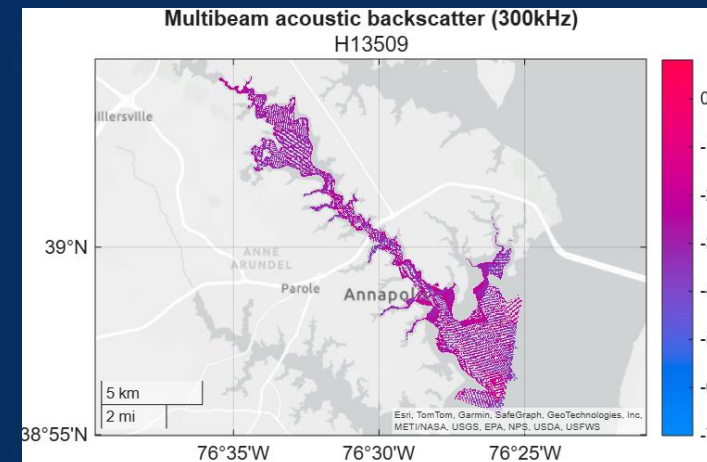
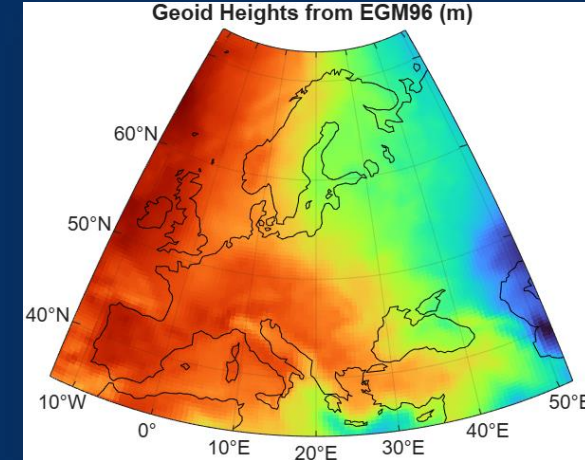


```
[A,R] = readgeoraster(filename);
```

New and Updated Map Visualizations – 2026a

New Functionality

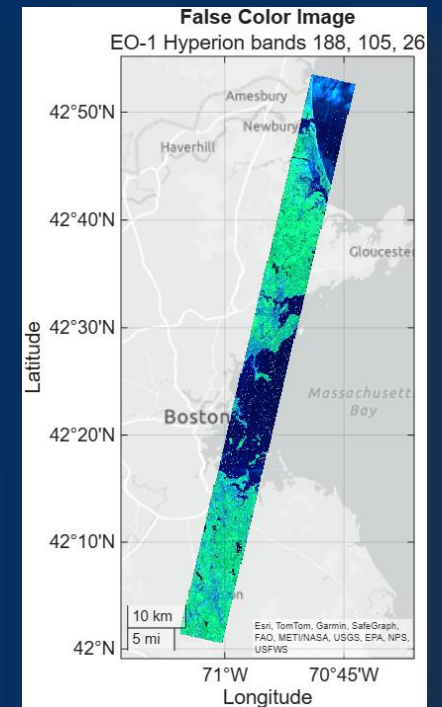
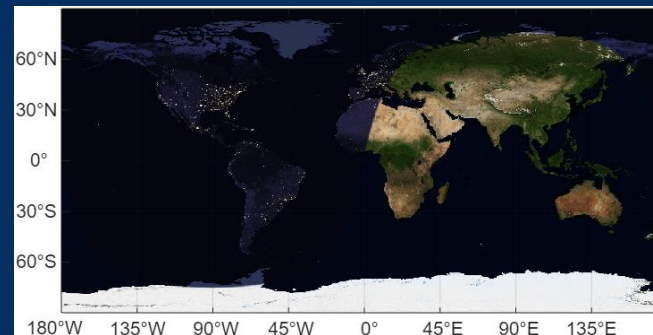
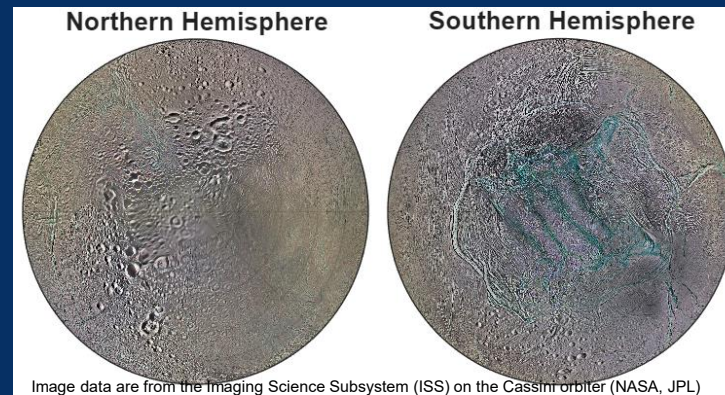
- Psuedocolor raster displays on maps



New and Updated Map Visualizations – 2026a

New Functionality

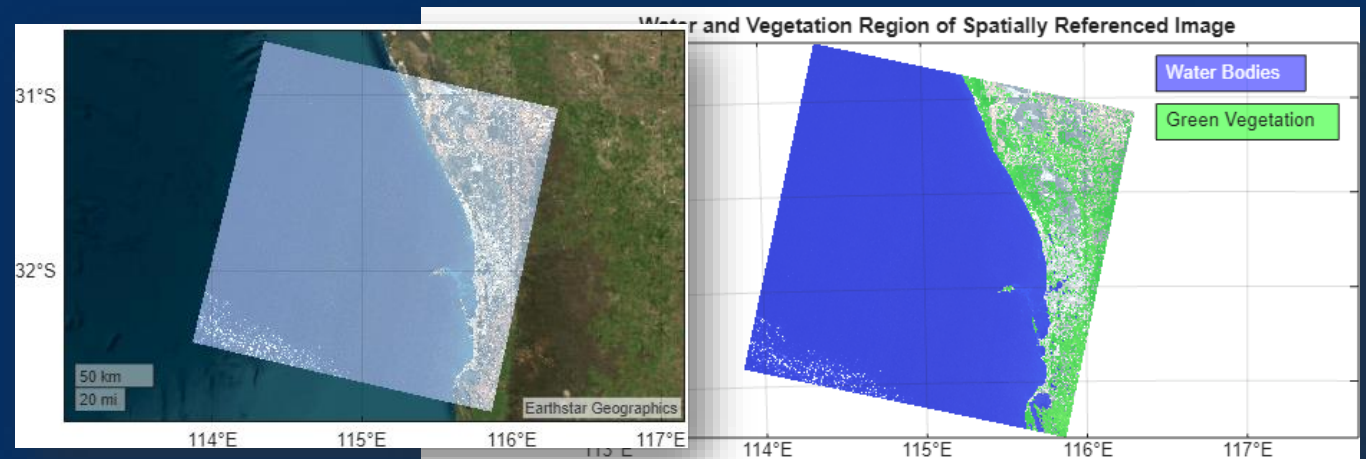
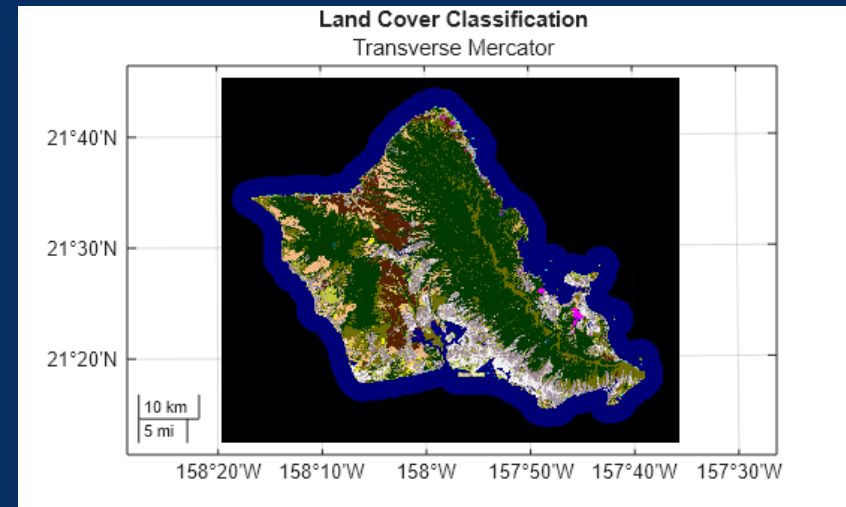
- Psuedocolor raster displays on maps
- Image displays on maps



New and Updated Map Visualizations – 2026a

New Functionality

- Psuedocolor raster displays on maps
- Image displays on maps
- Images and psuedocolor on map at same time
- Transparency effects on map layers

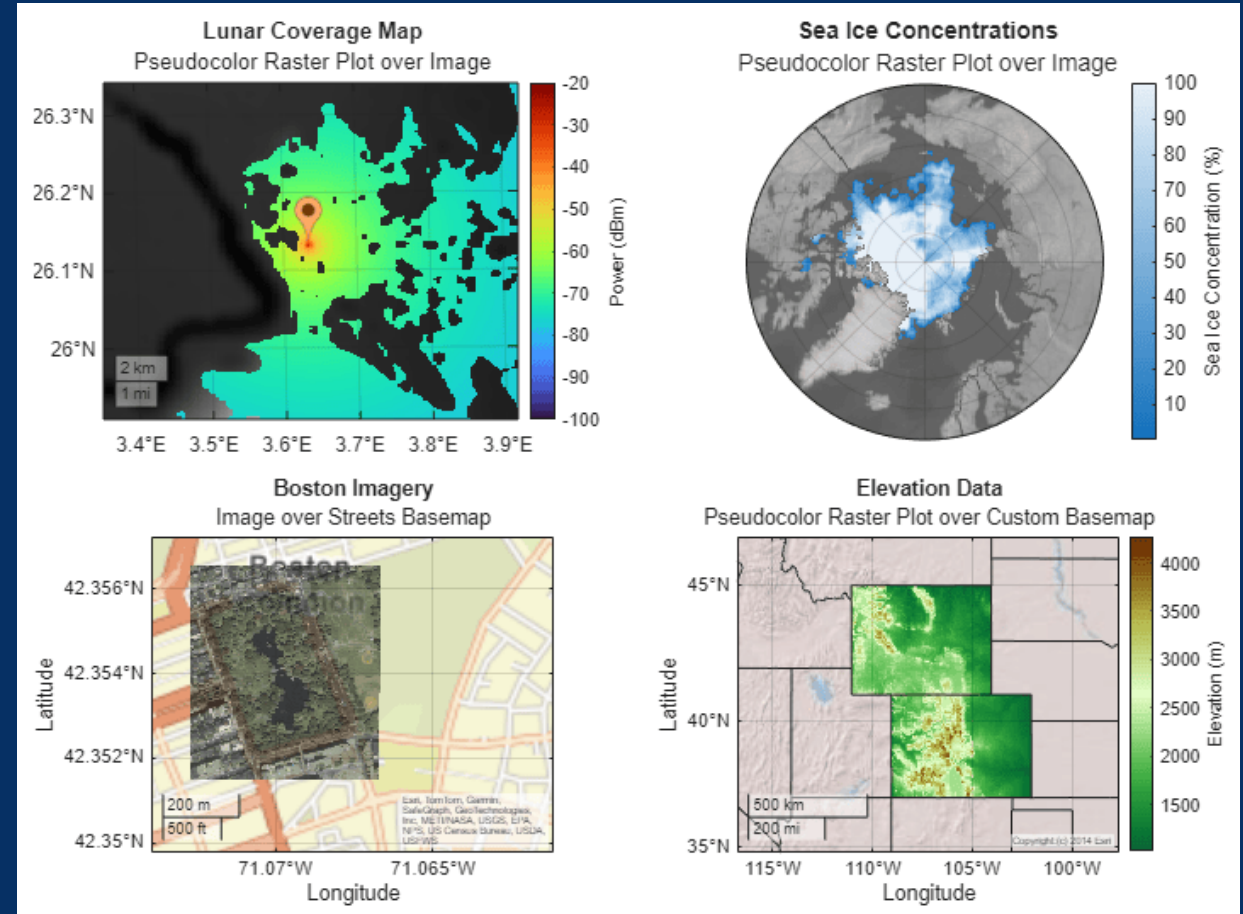


New and Updated Map Visualizations

Demo

Quality of Life Improvements – 2026a

- Map across international dateline/meridians
- Change projections more easily
- Redraw axes upon zoom
- Higher resolution/quality plots
- Merge multiple rasters or resample rasters
- Support for point and cell rasters

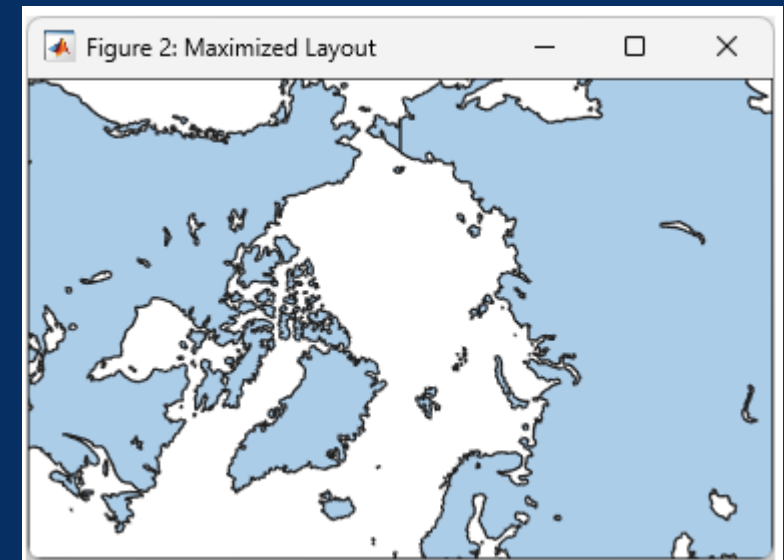
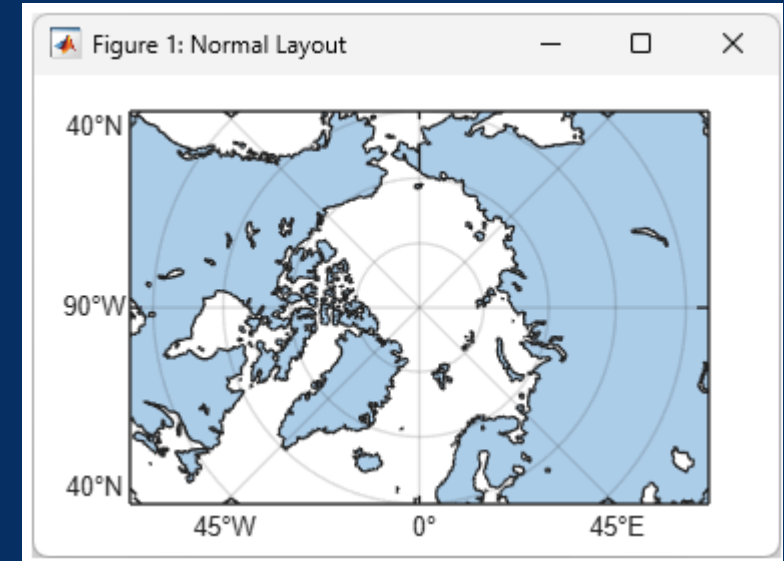


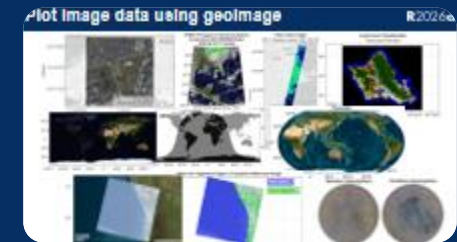
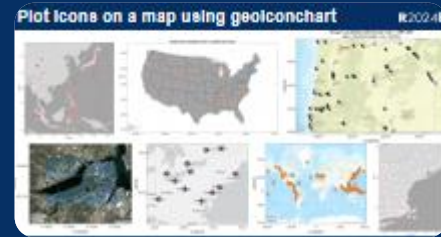
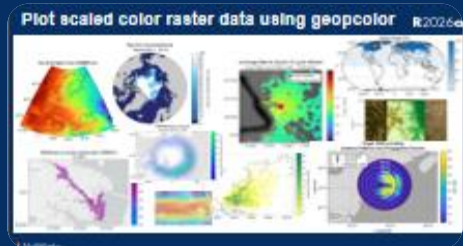
New and Updated Map Visualizations

Demo

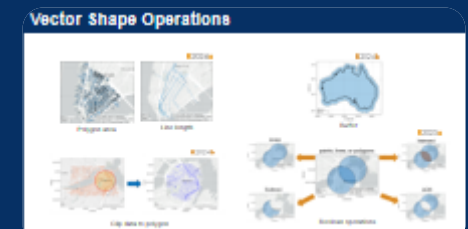
New Functionality & Quality of Life Improvements – 2026b

- Create map axes from placename or Aol
- Maximize map axes to fill space
- 3D geographic plots: lines, markers, tooltips updated
- CRS Handling
 - Native 3D coordinate support
 - Specify CRS of raster reference objects and shape objects on creation
 - Change/update CRS of plot





Limitless Library of Inspiring Ideas



Where to start?

Mapping Toolbox
Documentation & Examples

Theme - Integrations



Ease of Use



Software Workflows

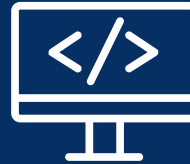


Integrations

Three topics for integrations



Languages



**Development
Environments**



AI Agents

Three topics for integrations



Languages



**Development
Environments**



AI Agents

Collaborate using seven bidirectional interfaces

Co-execute MATLAB with other languages and applications

Call code and libraries written in another language from MATLAB

Bidirectional interfaces for:

- Python
- Java
- C/C++
- .NET
- REST
- COM
- Fortran



» Call MATLAB code from another language



Example: calling python from MATLAB

- Run Python commands/scripts from MATLAB
 - py.*** - Execute a Python command inline: `py.max(3,4)`
 - pyrun** - Execute Python statements defined as strings
 - pyrunfile** - Execute a Python script file
 - Run Python Code** Live Task - Automatically generate MATLAB code to call Python
- Interchange Python datatypes within MATLAB
 - Numeric, string, list, tuple, datetime, timedelta, dictionaries, Pandas dataframe, ...
- Control MATLAB from Python
 - See [MATLAB Engine for Python](#) documentation

The screenshot displays the MATLAB Live Editor interface. At the top, a MATLAB script (line 1) defines a table `t` with 3 rows and 2 columns, `Var1` and `Var2`. The data is as follows:

	Var1	Var2
1	"a"	03-Sep-2024 12:35:19
2	"b"	04-Sep-2024 12:35:19
3	"c"	05-Sep-2024 12:35:19

Below the script, the **Run Python Code** task is configured. The input type is set to **Code**. The Python code entered is:

```
df = t
import pandas as pd
def add_column(df):
    df['new_col'] = (1,2,3)
    return df
add_column(df)
```

The output options are set to **Show code**. The result of the Python code execution is shown at the bottom (line 27), where the MATLAB `table(df)` command is used to convert the Pandas dataframe back to a MATLAB table. The resulting table `ans` has 3 rows and 3 columns, including the new `new_col`:

	Var1	Var2	new_col
1	"a"	03-Sep-2024 12:35:19	1
2	"b"	04-Sep-2024 12:35:19	2
3	"c"	05-Sep-2024 12:35:19	3

Manage python environments

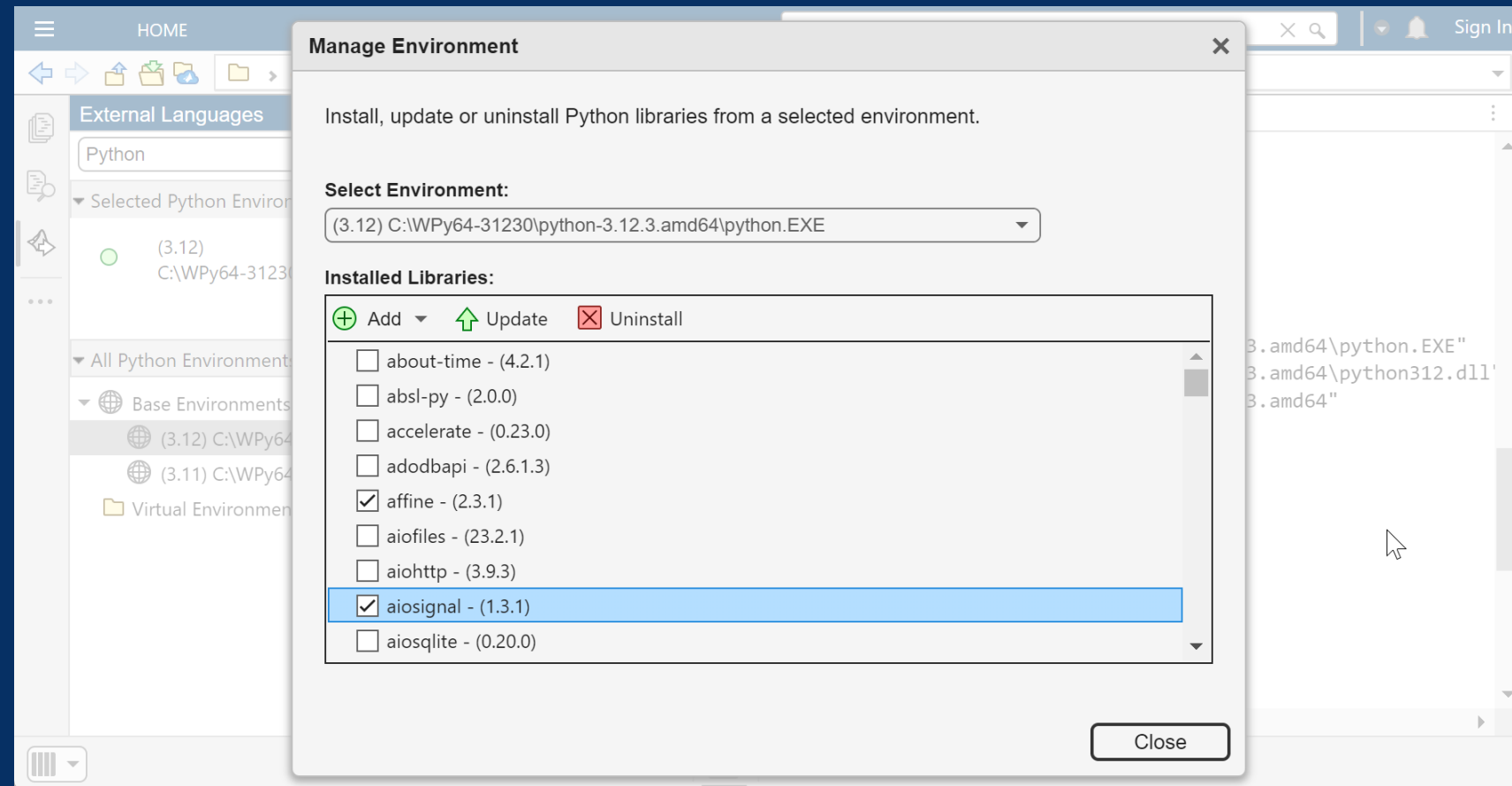
R2026a

Create and configure Python environments in the MATLAB desktop

Add or create new environments

Manage existing environments

Install, update, or uninstall libraries



Try it out!

Demo

Running Python with MATLAB

Table of Contents

Call Python from MATLAB

Option 1: Setup a Python Environment

Converting Data Types

Option 2: Use the Python Live Task

Export MATLAB as...

Compile MATLAB Code as a Standalone Executable or as Python Code

Call MATLAB from Python

Call Python from MATLAB

Option 1: Setup a Python Environment

Check which version of Python is installed and that it is on your MATLAB path. If running on MATLAB online, a version of Python is already installed on the MathWorks' servers.

If multiple versions of Python are detected or installed on your computer, specify the version of Python you want to run:

If you downloaded a Python interpreter, but did not register it in the Windows registry, use:

```
pyenv(Version="[executable]") % where [executable] is the path where the int
```

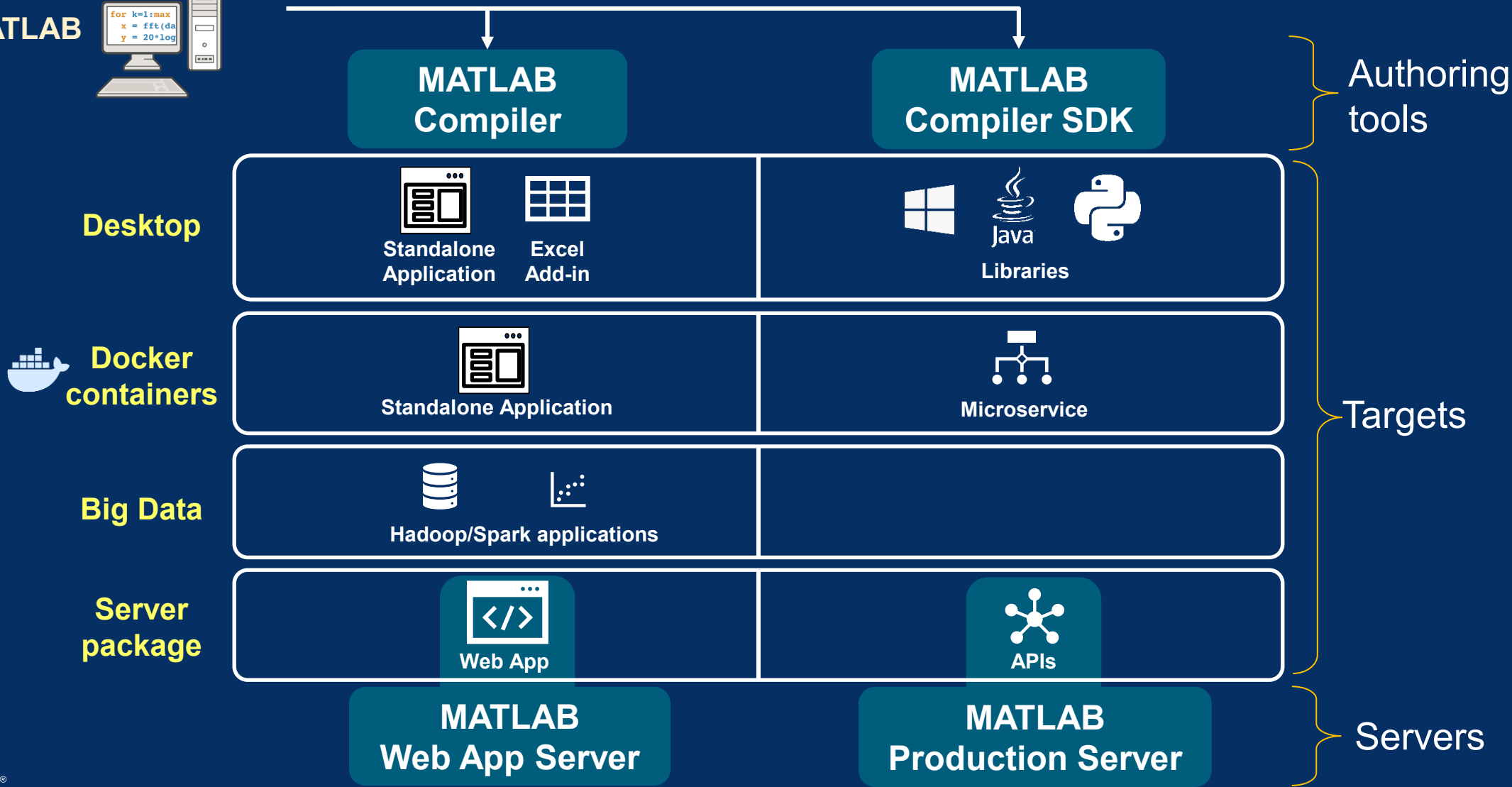
Call Python programmatically from
MATLAB

OR

Use a Python-MATLAB Live Task

Deploy MATLAB apps and APIs to non-MATLAB users

MATLAB

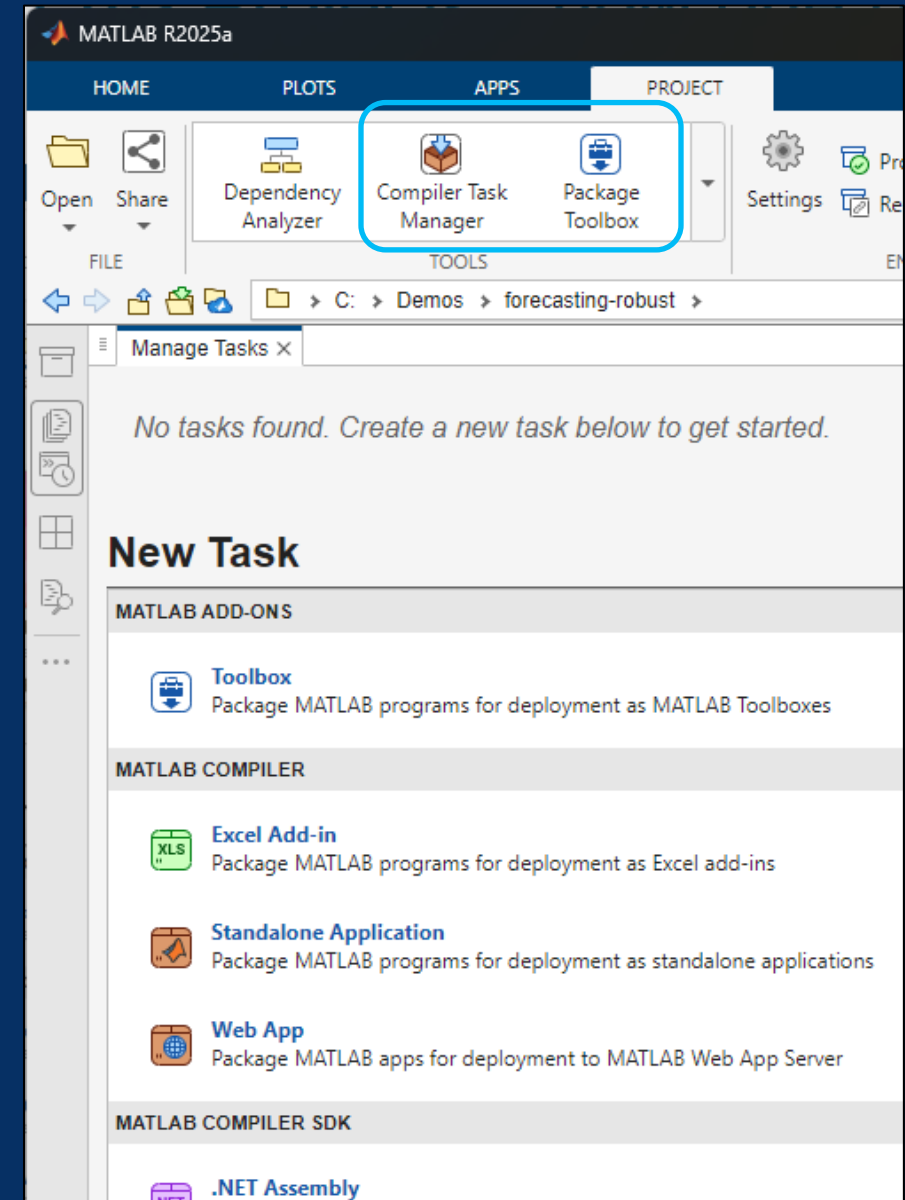


Deployment workflows have been streamlined

App, toolbox, and MATLAB Compiler workflows have been combined

New workflow uses MATLAB Projects

Dedicated interface for each deployment target



Integrate with AI Agents



Languages

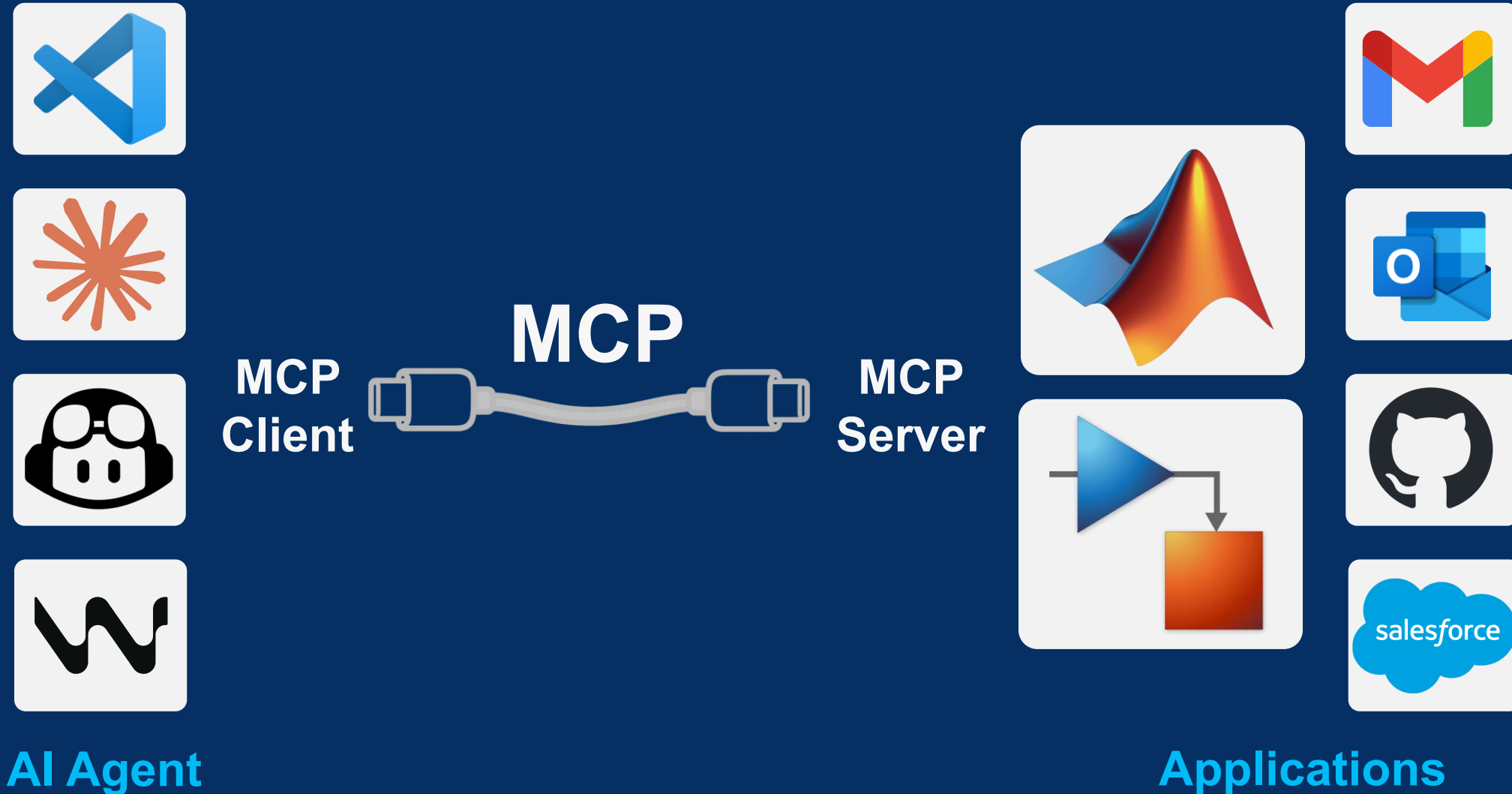


Development
Environments



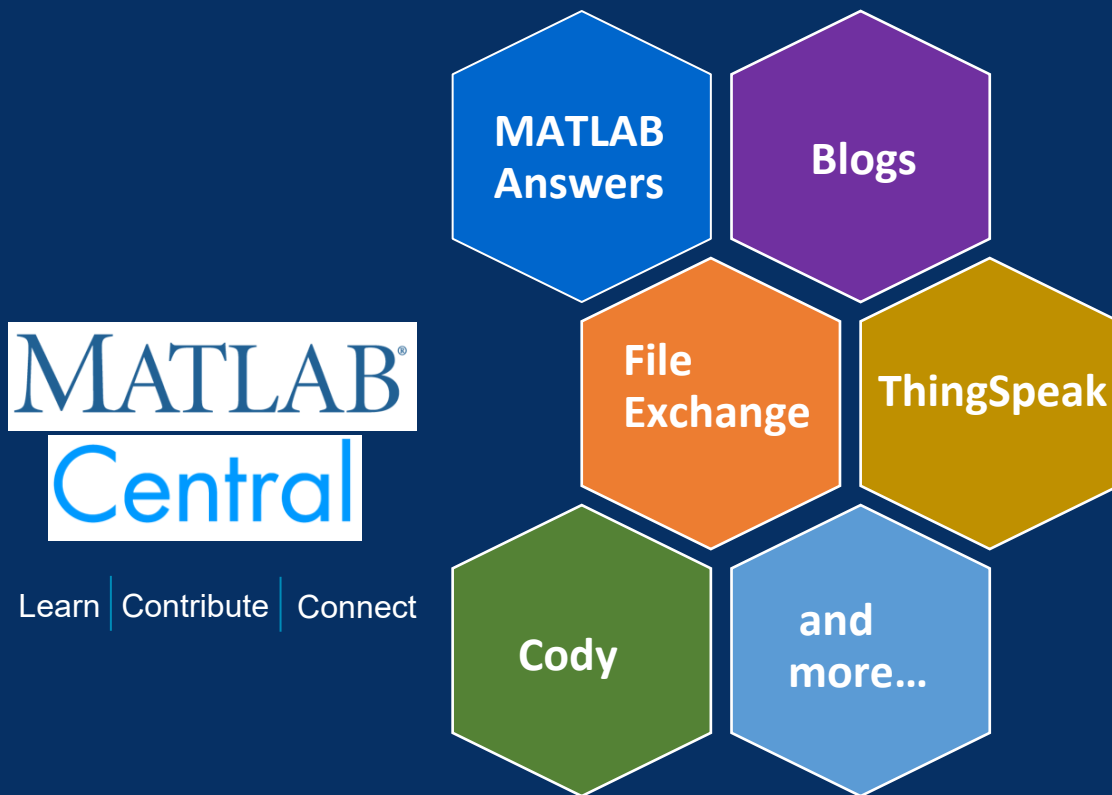
AI Agents

Enabled by Model Context Protocol (MCP)



MATLAB Central Community

Every month, over **2 million** MATLAB & Simulink users visit MATLAB Central to get questions answered, download code and improve programming skills.



MATLAB Answers: Q&A forum; most questions answered in < **1 hour**

File Exchange: Download code from a huge repository of free code including **tens of thousands** of open source community files

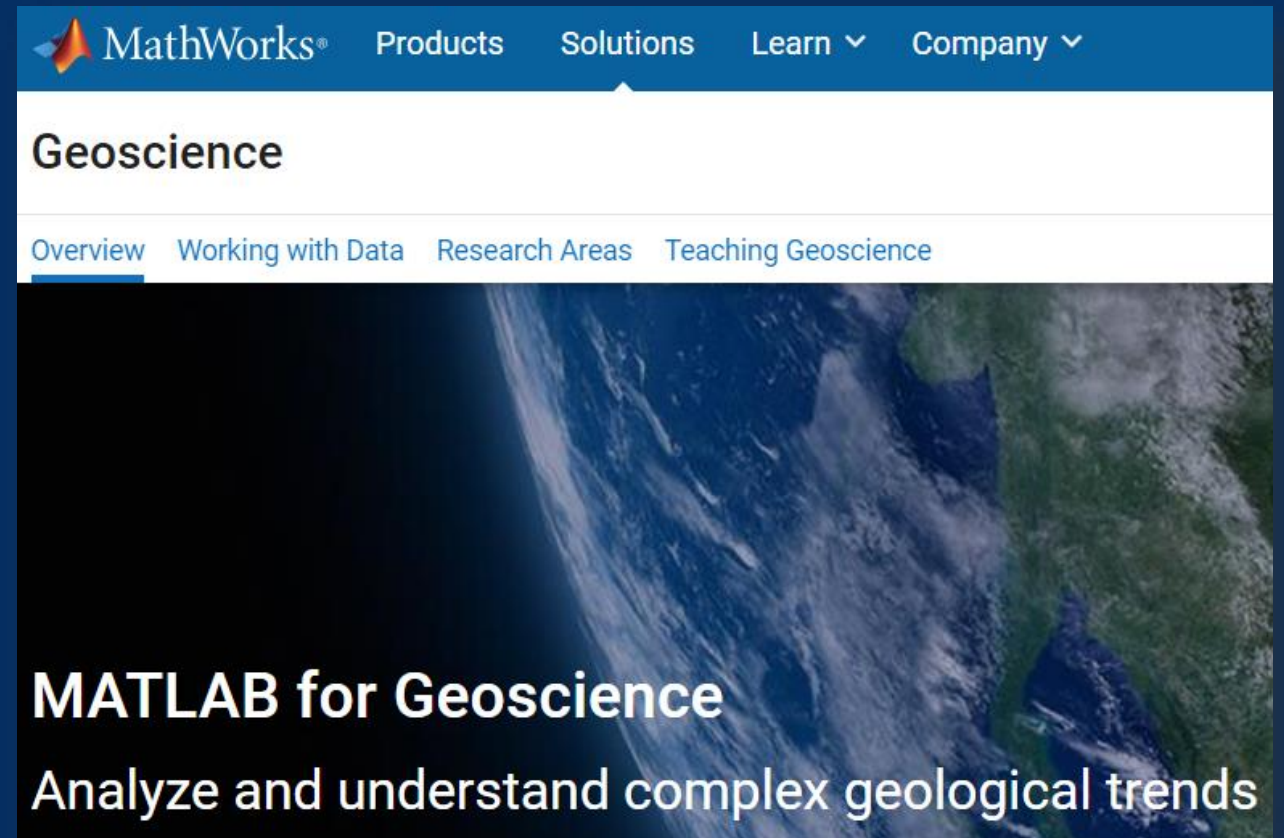
Cody: Sharpen programming skills while having fun

Blogs: Get the inside view from Engineers who build and support MATLAB & Simulink

ThingSpeak: Explore LoT Data

Check out the MATLAB for Geoscience webpage

- [Geoscience Overview](#)
- [Working with Geoscience Data](#)
- [Research Areas](#)
 - Seismology and Geophysics
 - Ocean and Climate
 - Hydrology and Water Science
 - Agriculture
- [Teaching Geoscience](#)





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